

A Framework for AI-Assisted Storytelling from Collective Multimedia Records

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Agenda

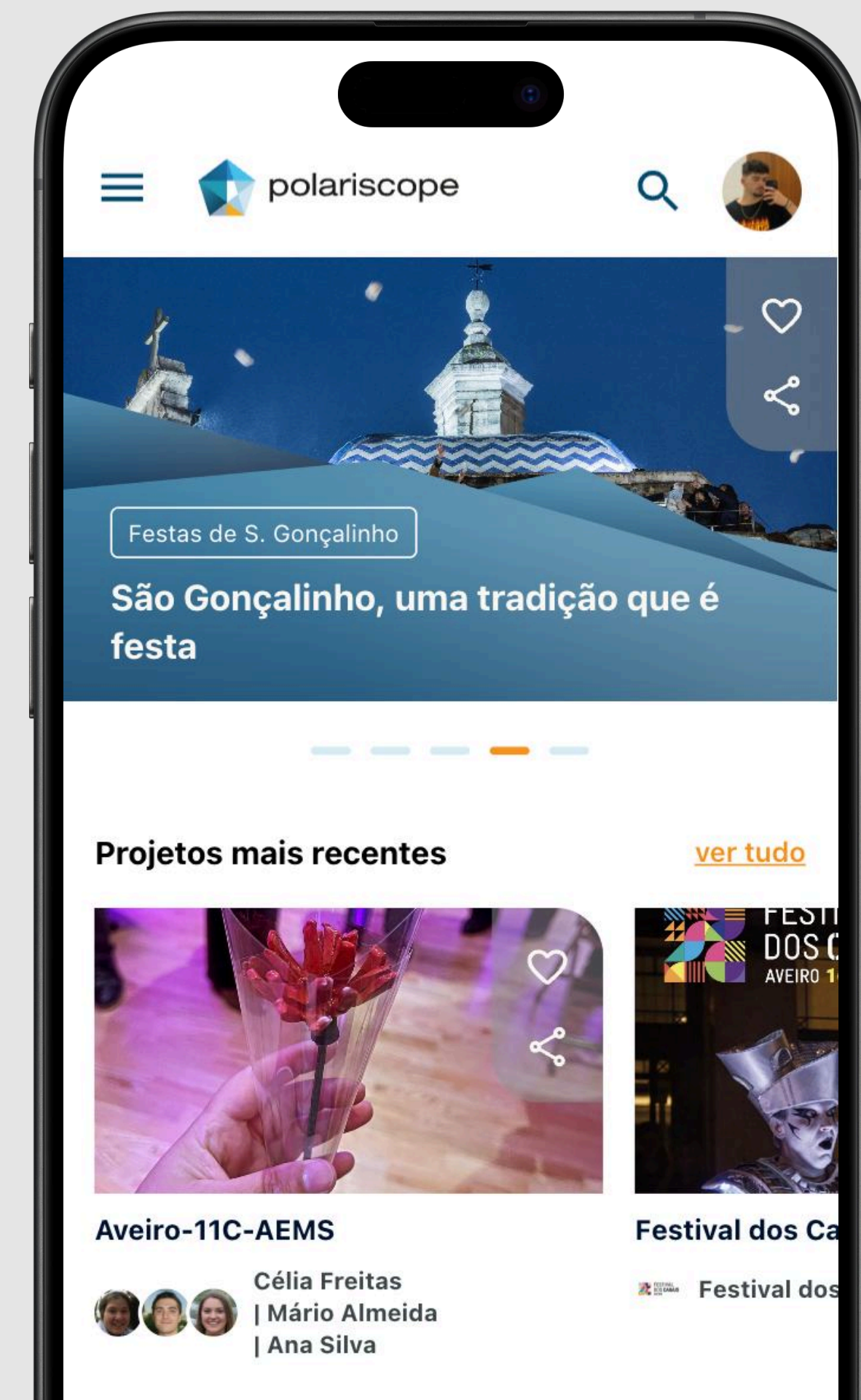
- I. Introduction
- II. The Problem
- III. Research question and goals
- IV. Research Structure
- V. Previous Work
- VI. Benchmarking
- VII. Proposed solution

Introduction

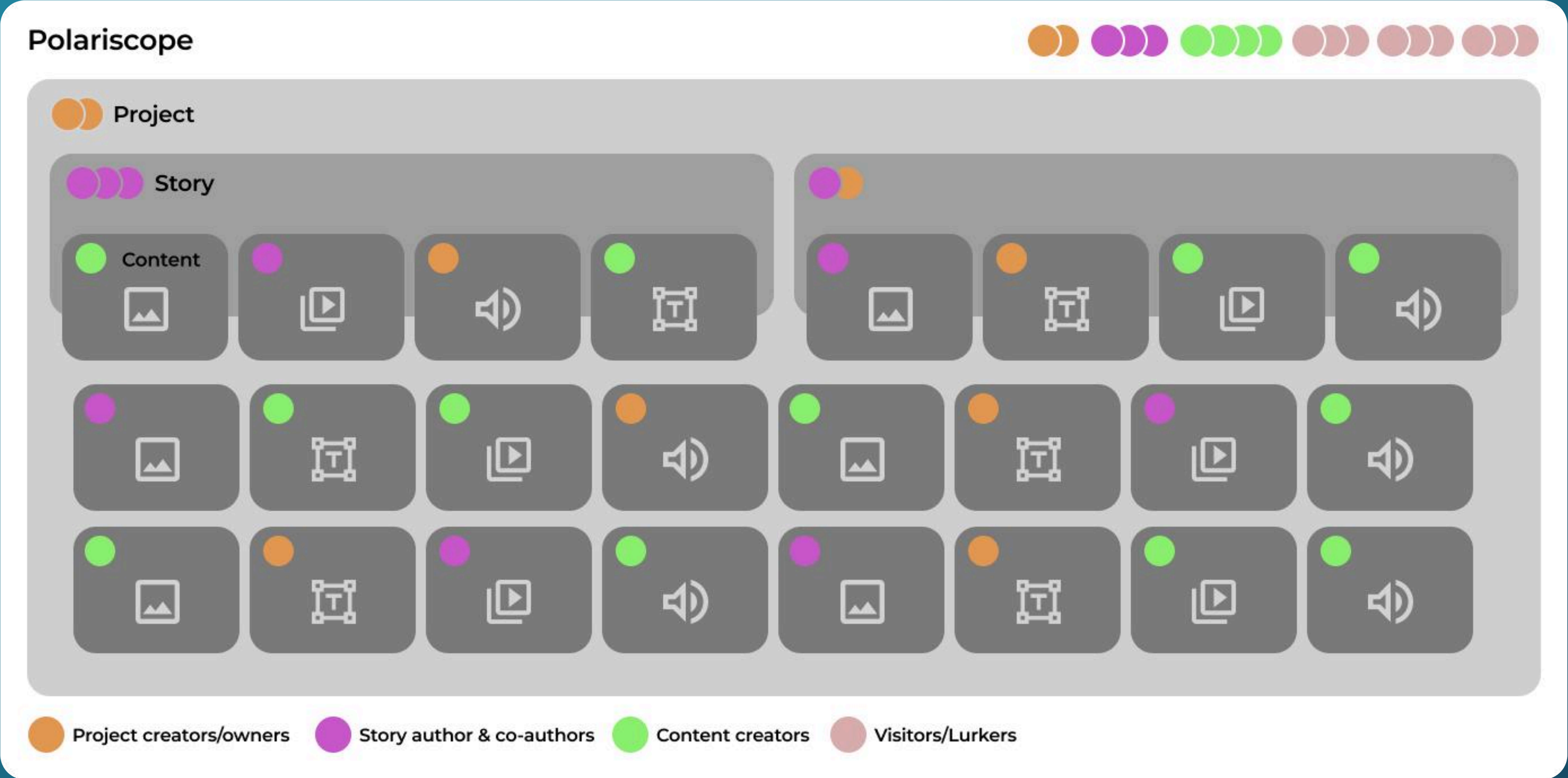
Polariscope is a **R&D project** to develop a collaborative storytelling platform.

The goal is to reunite citizens and cultural patrimony to share archives and memories promoting **cocreation and visualization**.

Its a solution that aims to assist users through **AI** in the process of **creating stories** and **generate stories** based on contents.



Introduction



The Problem

Most users prefer to **only post contents** or just see other people's contents.

Creating a story is a **long process** that requires a lot of attention, patience and creativity.

Our opportunity:

- Use **generative AI** to **assist story creators** by recommending contents, help writing and even fully automated creation of stories

Research Question and Goals

Can a generative AI assistant significantly contribute to the storytelling cocreation process and its perceived quality?



Research Question and Goals

- Develop a tool within the Polariscope platform to **facilitate and enhance the story cocreation process** through the integration of AI
- Specify functional and interaction requirements for the assistant's implementation
- Develop an **AI assistant** capable of **generating stories** from existing content while ensuring creative control by the users
- Evaluate, based on story creators' perception, the **quality and satisfaction** of the support provided by the assistant on the platform
- Identify users' control expectations when utilising the assistant

Research Structure

My research method is a **Design Based Research**:

- Focused on **helping the story creators**, not only on making the feature work

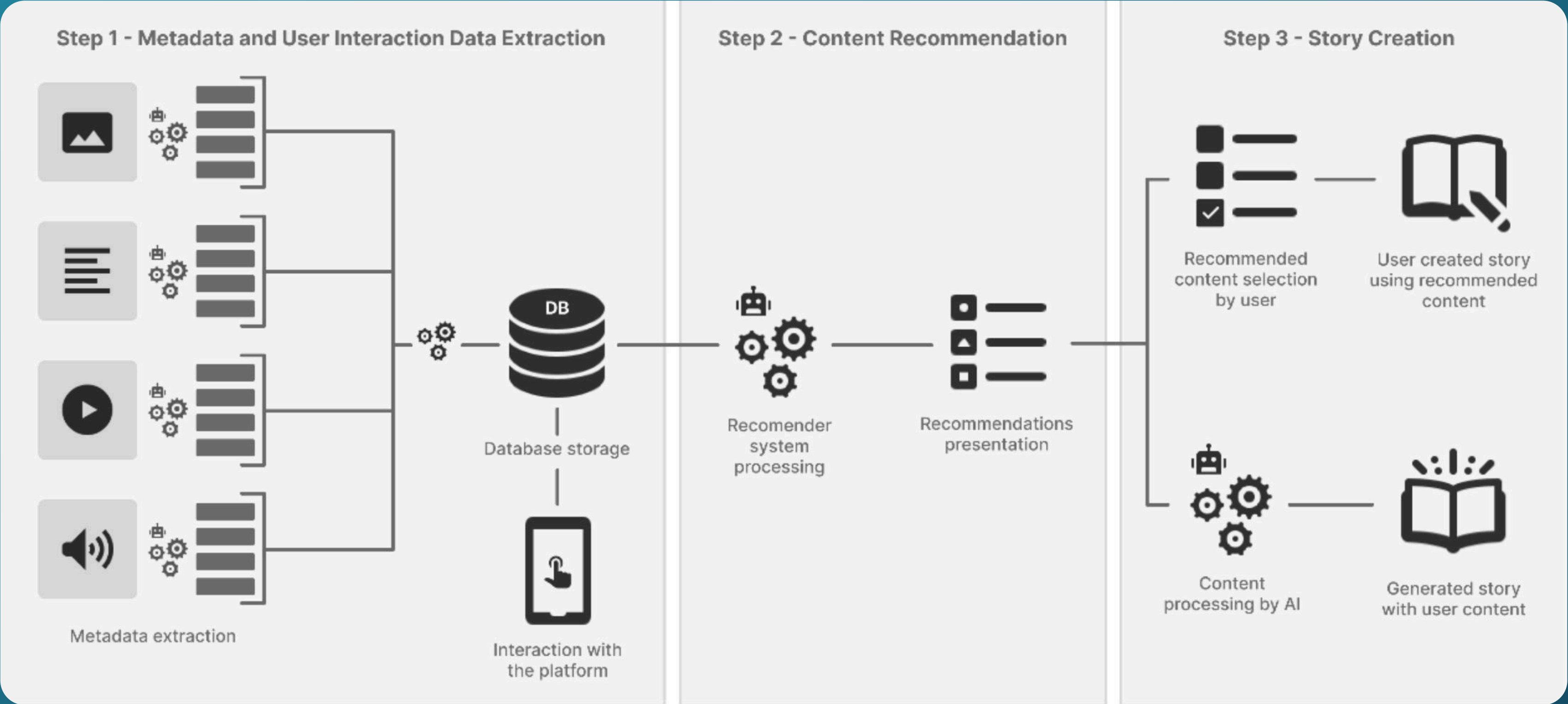
Goal is to make the process of creating stories **easier and better**.

Make it a **good and intuitive experience**.

Research Stages

1. Benchmarking
2. Proposed Solution
3. Development
4. Interactive Evaluation
5. Final Adjustments

Previous Work



Previous Work

Tests were conducted with ChatGPT 3.5.

The **AI's Temperature** parameter cannot be a fixed value for heritage storytelling.

The preferred Temperature depends entirely on the **story's context** and the required level of **factuality**.



AI Benchmarking

Temperature, **Top-p** and **Repetition penalty** are the chosed parameters.

ChatGPT and **Gemini** as main options. Excellent LLMs with good APIs.

Open source models like **Llama 3** also have advantages but will have to be better evaluated because of the hardware cost to maintain.

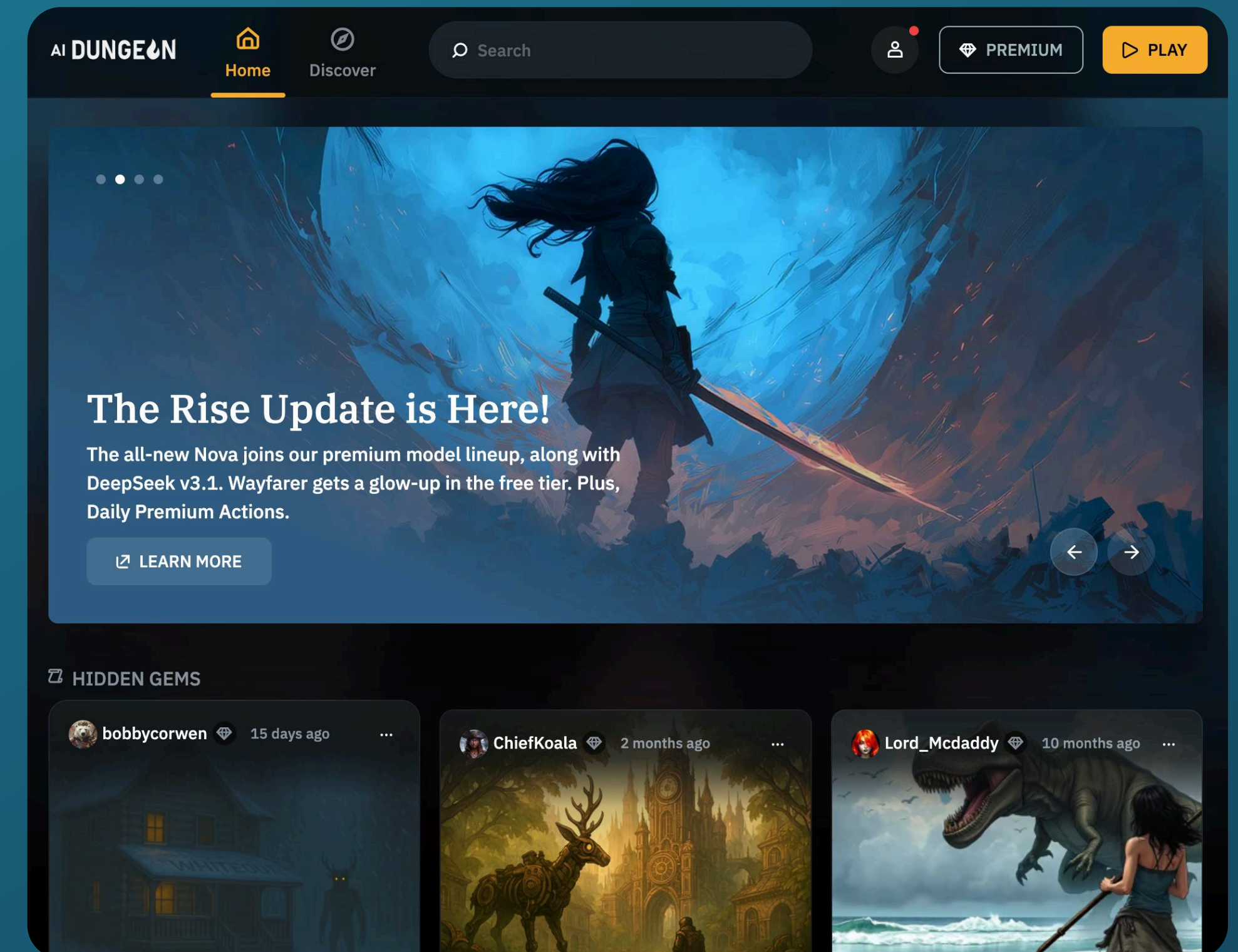
Other models like **Claude 3** and **Deepseek** were explored but we don't think they bring benefits further than Gemini for example.

Platform Benchmarking

- AI Dungeon, NovelAI and Artflow
- Sudowrite, Grammarly and Canva Magic Tool

Confirmed the previous work results.

Decided to go with **integrating AI APIs** directly in the platform.



aidungeon.com

Proposed Solution

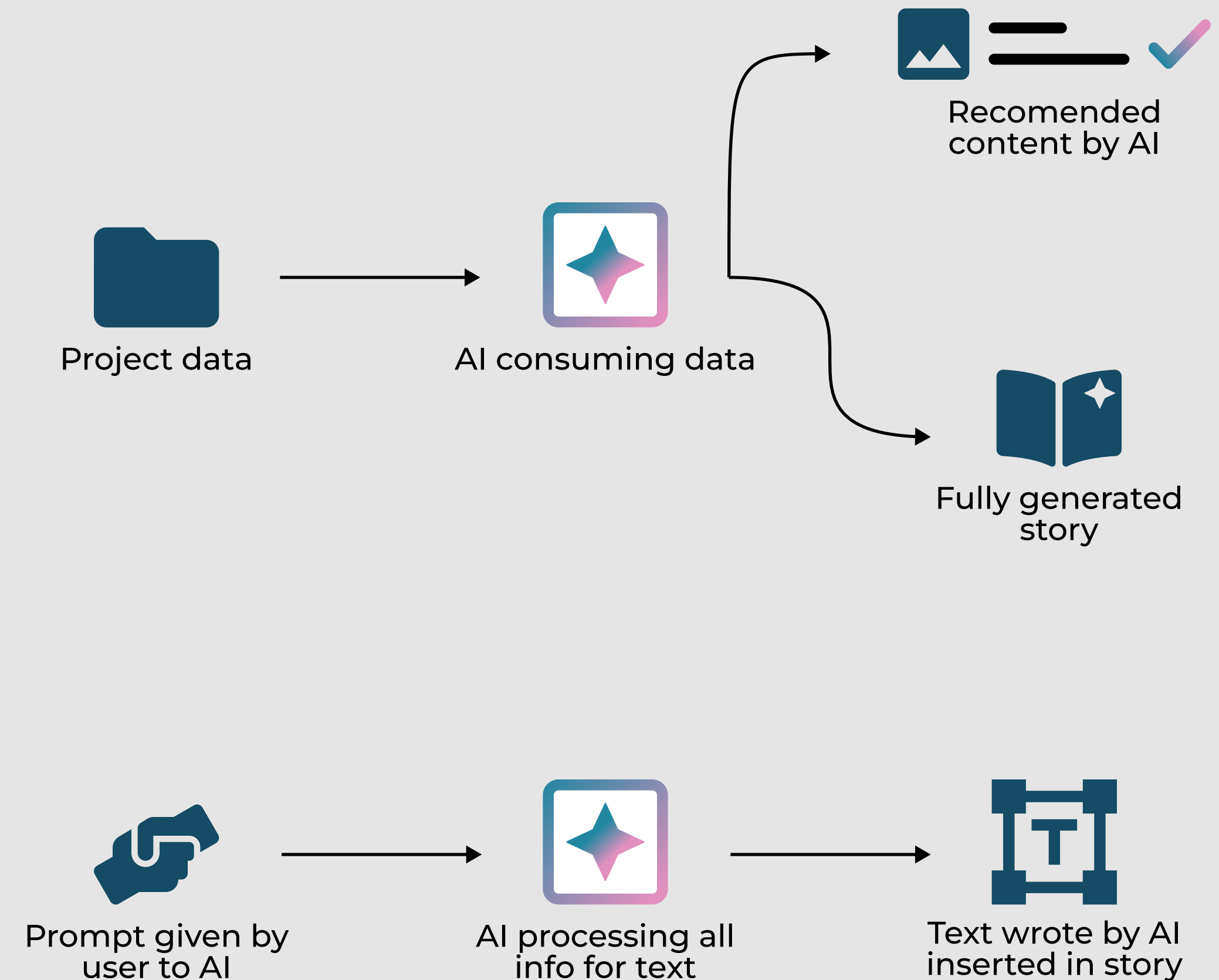
Focused on assisting on the correct spot only, through discret and **intuitive button** placements that open the AI assistant to **help the creators**.



PolarisAI

Proposed Solution

- **Recommendation of content** that already exists in the Project
- Fully automatic **creation of stories** with the Project data
- **Help writing** and rewriting blocks of text



Proposed Solution

With a simple prompt PolarisAI can:

- Write the story **title**
- Write the story **description**
- **Correct** your text
- Write a **new block of text**

The screenshot displays the Polariscope web application interface. The top navigation bar includes the Polariscope logo, links for 'Projetos', 'Histórias', 'Sobre', 'Ajuda', and 'Contactos', a search icon, and an 'Entrar' button. The left sidebar contains three main steps: '1. Detalhes', '2. Conteúdos' (highlighted), and '3. Rever e Publicar'. The '2. Conteúdos' step is further divided into 'PolarisAI' and 'Escreve aqui'. The 'PolarisAI' section prompts the user with 'Queres ajuda a escrever a tua história?' and features a text input field. The 'Escreve aqui' section is a large text area for writing. The main content area shows a project title field with a placeholder 'Este título pode ter até 3 linhas ou cerca de 70 caracteres no máximo.' Below this, there are sections for 'Criadores' (listing 'Mordomia S. Gonçalves' and 'Universidade de Aveiro') and 'Sobre esta história' (providing a detailed description of the story). At the bottom, there are four keyword tags: 'keyword_1', 'keyword_2', 'keyword_3', and 'keyword_4'. A 'story_divider' section is also visible, containing a block of text about the 25th of April revolution in Lisbon.

polariscope Projetos Histórias Sobre Ajuda Contactos

1. Detalhes

2. Conteúdos

3. Rever e Publicar

PolarisAI

Queres ajuda a escrever a tua história?

Escreve aqui

Nome do projeto

Este título pode ter até 3 linhas ou cerca de 70 caracteres no máximo.

Criadores

Mordomia S. Gonçalves

Universidade de Aveiro

Sobre esta história

Escrevo esta história para que a memória do 25 de abril nunca seja esquecida por todos nós. Na tarde do dia 28 de abril, sob o sol radiante, as ruas de Lisboa vibravam com a chegada triunfante de Mário Soares, aclamado por centenas em êxtase.

keyword_1 keyword_2 keyword_3 keyword_4

Tendo presente a revolução dos cravos, faz-se uma viagem pelo passado, relembrando momentos marcantes. Iniciam no dia histórico em que os militares comandados por Salgueiro Maia ocupam o Terreiro do Paço em Lisboa. Um tanque e militares armados protagonizam a cena com a esperança de um novo dia desponta na mente de alguém.

story_divider

Tendo presente a revolução dos cravos, faz-se uma viagem pelo passado, relembrando momentos



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Thank You

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